

Juan Montesinos Benzal

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[Engineering Portfolio](#)

EDUCATION

University of Bath

BEng Mechanical Engineering; Year 1 average: 82%; Chancellor's Scholar Sep. 2025 – Jun. 2029 (expected) Bath, UK

- Design, Materials & Manufacturing 1 (73), Engineering Mathematics (89), Thermofluids 1 (82), Mechatronics (84), Solid Mechanics 1 (85).

The Global College

International Baccalaureate Diploma, 40/45 Sep. 2023 – Jun. 2025 Madrid, Spain

- HL: Computer Science (6), Mathematics AA (6), Physics (7). SL: Business (6), Spanish (6), French (7).

EXPERIENCE

Engineering Intern

Jun. 2026 – Sep. 2026

Automotiva *On-site*

- Supported Balance of Performance engineering across TCR Spain, International GT Open and GT Cup Europe, configuring data loggers and handling trackside data acquisition and retrieval.
- Analysed vehicle performance data and assisted data scrutineering, checking recorded datasets and supporting senior engineers during race events.

Technical Advisor

Nov. 2024 – Present

STEM Racing Spain *Madrid, Spain*

- Support regional and national competitions through technical scrutineering, rule interpretation and engineering guidance; teach CAD, CFD, FEA and CAM principles to competing teams.
- Develop the automated timetable-generator app and collaborate on the company database and customer-facing competition app, supporting scheduling, team data and competition workflows.
- Co-designed an STM32-based miniature vehicle data logger with onboard storage, accelerometer, gyroscope and magnetometer, plus custom analysis software inspired by Cosworth Pi Toolbox.
- Support competition-car manufacturing and operate a 4-axis CNC machine.

Coding Teacher

Oct. 2023 – Mar. 2025

CodeForAll Madrid *Madrid, Spain*

- Volunteer teaching Python and Java to school students from underprivileged families without access to paid coding education, using practical projects to develop programming and digital-literacy skills.

ENGINEERING PROJECTS

Team Bath Zero MotoStudent

Jan. 2026 – Present

Chassis Engineer

- Optimised a tubular motorcycle chassis using 2D parametric and 3D FEA in Ansys Mechanical; performed fatigue and modal analyses to evaluate durability and dynamic behaviour.
- Designing and validating a topology-optimised triple clamp for Laser Powder Bed Fusion (LPBF), defining representative load cases and performing FEA and modal analysis.
- Completed the mechanical design of an electric-motor dynamometer test rig, defining the test layout, mounting strategy and component interfaces.

Formula Student | Team Bath Racing Electric

Oct. 2026 – Present

Simulation Engineer

- Designed and validated a mass-optimised anti-submarine belt fixture for the driver restraint system.
- Designed and validated a mass-optimised brake pedal while meeting structural requirements.
- Designed a composite front-hatch access cover using surface modelling, developing the geometry for integration with the monocoque.

F1 in Schools | Velotech

Sep. 2023 – Dec. 2024

Team Leader & Designer

- Led vehicle design and validation using CAD, CFD and FEA, progressing the final design through CNC manufacturing for competition.
- Finished 2nd at the 2024 Spanish National Finals, qualifying to represent Spain at the World Finals in Saudi Arabia; nominated for the Chair of Judges Recognition of Achievement Award.

Enlil1 | CanSat

Sep. 2022 – May 2023

Team Leader & Head of Design

- Led mechanical design and developed ESP32-based communications using LoRa, MQTT and Bluetooth for the CanSat mission.
- Placed 3rd in the Madrid regional competition and received the Best Technical Development award.

SKILLS

CAD: Inventor, Fusion 360

CAE: Ansys Mechanical, Fusion 360 CAM

Programming: Python, C++, MATLAB, Arduino

Manufacturing: 4-axis CNC, composite manufacturing

Languages: English, Spanish